Kieran Choi-Slattery

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Education

PhD **Georgia Institute of Technology**, Electrical and Computer Engineering — Atlanta, GA
- GPA: 3.75/4.00

Aug 2025 – present

BS **Georgia Institute of Technology**, Aerospace Engineering — Atlanta, GA
- GPA: 3.50/4.0
- Minor: Earth and Atmospheric Sciences, Geophysics Track

Aug 2021 – May 2025

Experience

Georgia Tech Planetary Exploration Laboratory, Graduate Research Assistant — Atlanta, GA

Aug 2023 – present (3 years 3 months)

- Developed inversion algorithm to jointly estimate Venus cloud particle sizes and refractive index from single-particle light scattering.
- Developed optical setup for algorithm validation with laser alignment, spatial filter, and dual-polarized collection optics.
- Designed, built and tested custom programmable-gain photodiode amplifier PCBs achieving sub-nA sensitivity over 45 kHz bandwidth.

Georgia Tech Glacial Geophysics Laboratory, Undergraduate Research Assistant — Atlanta, GA

Aug 2022 – May 2024 (1 year 10 months)

- Implemented digital beamforming algorithms on ice-penetrating MIMO radar array.
- Used qGIS to determine optimal transect locations for bistatic radar tomography experiment in Helheim Glacier, Greenland.
- Used RF test equipment to troubleshoot radar echo sounder units for field deployment.

Projects

Flight Computer for Georgia Tech Experimental Rocketry — Jan 2022 – May 2025

Flight computer for a high-altitude 2-stage sounding rocket.
- As Avionics Lead, led development of 5-board custom flight computer commanding second stage ignition, state estimation, and parachute deployment.
- Wrote firmware in C for STM32 microcontrollers.
- Designed mounting hardware using fiberglass, aluminum, and 3-D printed materials.

Tri-band FMCW MIMO Radar — Feb 2026 – present

8x8 sparse radar array operating at 2.4, 5.6, and 10 GHz made from fully custom PCBs.
- Optimized array layout for beamforming based on convex programming.
- Designed transmitter module, including PLL+Synth, power amplification, and switches.

Publications

Development of a data retrieval algorithm for Venus nephelometry in preparation for the Rocket Lab Mission to Venus — Nov 2026

M. Regina A. Moreno, Christopher E. Carr, Darrel Baumgardner, *Kieran Choi-Slattery*, Snigdha Nellutla, Iaroslav Iakubivskyi, Kevin T. Jansen, Janusz J. Petkowski, Sara Seager
10.1016/j.icarus.2026.117189 (Icarus)

Skills

CAD Software: Altium, KiCAD, SolidWorks

Programming Languages: Python, MATLAB, C, C++, Java, JavaScript/TypeScript

Research Areas: Space Instrumentation for Planetary Science, Inverse Problems